

Firearm Carrier Detection and Threat Assessment Using Multi-Model Deep Learning

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Abstract

The increasing prevalence of firearm-related incidents in public spaces necessitates automated surveillance systems capable of real-time threat identification. This paper presents a multi-model deep learning pipeline for detecting and classifying human-firearm interactions from single images. Our system integrates (1) a YOLOv8 object detection stage for localizing people and firearms, (2) a three-stream ResNet-50 CNN for classifying interactions into carrying, holding, or aiming categories, (3) a BLIP scene captioning module for generating natural-language scene descriptions, and (4) a DistilBERT-based NLP module for semantic threat analysis. We achieve 97.5% validation accuracy on HOI classification and 0.800 mAP@0.5 on object detection using the LFC dataset. The system produces an interpretable fused threat score combining visual and textual signals, enabling LOW, MEDIUM, and HIGH threat assessments. Our results demonstrate the effectiveness of combining multiple visual and language models for automated threat assessment in surveillance applications.

1. Introduction

The rising frequency of firearm-related incidents in public spaces has created an urgent need for automated surveillance systems capable of instantly identifying and assessing potential threats [1]. Traditional surveillance methods rely on human operators monitoring multiple camera feeds, a

process that is inherently labor-intensive, prone to fatigue, and error-prone. As highlighted by Mahmood et al. [1], the reliance on human observation for extended periods poses a risk of errors, as even a momentary lapse in concentration can lead to disastrous consequences. Deep learning offers a promising alternative through automated visual recognition of human-firearm interactions.

In this paper, we propose a multi-model deep learning pipeline for the identification and categorization of gun-related Human-Object Interactions (HOIs). Our system detects whether a person is carrying, holding, or aiming a firearm and quantifies the threat level for each detected interaction. The pipeline integrates four core components: (1) a YOLOv8 [2] detection stage to localize people and firearms, (2) a three-stream ResNet-50 CNN for HOI classification, (3) a BLIP scene captioning module that enriches threat scoring with natural-language context, and (4) a DistilBERT NLP module that performs text-based threat analysis on generated captions to produce a language-grounded threat score.

Recent advances in weapon detection have demonstrated the effectiveness of various deep learning approaches. Sharma and Arora [2] achieved 88.2% mAP using YOLOv8 for constructive weapon detection. Vijayakumar et al. [3] compared R-CNN and YOLOv4 architectures, achieving 96.04% mAP for real-time weapon detection. Yadav et al. [4] extended weapon detection to video surveillance with real-time handgun tracking capabilities. Torregrosa-Domínguez et al. [5] addressed the challenge of small

weapon detection using scale-matching techniques, achieving a 71% reduction in false positives.

The key contributions of this work include:

- A complete end-to-end pipeline for firearm-related HOI detection and threat assessment
- A three-stream ResNet-50 architecture that achieves 97.5% validation accuracy on interaction classification
- A multimodal threat scoring system that fuses visual detection, interaction classification, and language-based semantic reasoning
- Comprehensive evaluation on the LFC dataset with detailed ablation and error analysis

Our approach demonstrates that combining visual detection, interaction classification, and scene captioning yields an interpretable and robust system for automated threat assessment. The code and model checkpoints are available at: <https://github.com/msds25050-cpu/firearm-hoi-detection>

2. Related Work

2.1. Human-Object Interaction Detection

HOI detection has been extensively studied in computer vision. Early approaches employed two-stage detection pipelines with hand-crafted features, while recent methods leverage deep learning for end-to-end learning. Mahmood et al. [1] introduced a novel attention-based mechanism for localizing firearm carriers in complex scenes, leveraging human-firearm interaction information to provide valuable clues for carrier localization. Their approach incorporated an attention mechanism that effectively distinguishes humans and firearms from the background by focusing on relevant areas, achieving 77.8% AP on the LFC dataset.

2.2. Firearm Detection in Surveillance

Several works have specifically addressed firearm detection in surveillance contexts. Mahmood et al. [1] introduced the Localization of Firearm Carriers (LFC) dataset, consisting of 3,128 images with meticulous annotations for humans and firearms, including guns and rifles. Their work represents the first attempt to address the specific aspect of identifying firearm carriers in images.

Sharma and Arora [2] proposed the All-Seeing Eye system for constructive weapon detection using YOLOv8, achieving 88.2% mAP on a custom dataset of 16,000 weapon images. Their approach addressed the challenge of weapon detection in diverse environments, including various lighting conditions and backgrounds.

Vijayakumar et al. [3] developed a comprehensive framework combining R-CNN and YOLOv4 for intelligent weapon detection in real-time video. Their model achieved 96.04% mAP with YOLOv4, demonstrating the effectiveness of one-stage detectors for rapid threat identification.

Yadav et al. [4] extended weapon detection to video surveillance with real-time handgun detection and tracking capabilities. Their approach addressed the temporal dimension of threat assessment, enabling continuous monitoring of potential threats across video frames.

Torregrosa-Domínguez et al. [5] focused on enhancing real-time weapons detection through effective strategies including scale-matching techniques for small objects. Their approach achieved a +13.23 AP improvement for small object detection and a 71% reduction in false positives.

2.3. Multimodal Approaches

Recent research has explored combining visual and textual modalities for improved scene understanding. While most approaches focus purely on visual detection, our work introduces a novel combination of visual detection, interaction classification, and scene captioning to incorporate semantic context into threat assessment, distinguishing our system from purely vision-based methods.

2.4. Threat Scoring Systems

Existing threat assessment systems typically rely on binary classification or simple rule-based scoring. Our approach introduces a continuous, interpretable threat score that combines multiple signals, providing nuanced risk assessment suitable for security applications. This represents a significant advancement over traditional binary classification approaches to firearm threat assessment.

3. Data

3.1. LFC Dataset

Our project is based on the LFC (Localizing Firearm Carriers) dataset introduced by Mahmood et al. [1]. The dataset contains 3,128 annotated images with interaction labels and bounding boxes for both people and firearms. The dataset includes three interaction classes: carrying (a person holding a firearm in a non-threatening posture), holding (a person gripping a firearm in a ready position), and aiming (a person directing a firearm at a target). As noted by Mahmood et al. [1], the LFC dataset presents several challenges due to varying overlaps between humans and firearms, cluttered backgrounds, viewpoint variations, and complex interaction scenarios.

After loading the labeled interactions and collecting carrying images, the dataset produced 3,433 total annotations, as some images contain multiple person-firearm pairs. Following the standard split, we allocated 70% for training, 15% for validation, and 15% for testing as shown in Table 1.

3.2. Preprocessing and Augmentation

All images were resized to 224×224 pixels for CNN training and 640×640 pixels for YOLOv8 detection. For training, we applied random horizontal flipping, color jitter

Table 1. Dataset splits

Split	Fraction
Train	70%
Validation	15%
Test	15%

(brightness 0.3, contrast 0.3, saturation 0.2), and normalization with ImageNet mean [0.485, 0.456, 0.406] and standard deviation [0.229, 0.224, 0.225]. Validation and test sets were only resized and normalized.

Following the strategies outlined by Torregrosa-Domínguez et al. [5] for improving small object detection, we employed targeted augmentation techniques to enhance firearm detection performance. To address class imbalance, we computed inverse-frequency class weights and incorporated them into the CrossEntropyLoss function, assigning greater importance to minority classes during training.

4. Methods

4.1. System Architecture Overview

Our pipeline comprises four sequential stages:

1. **Detection:** YOLOv8 detects people and firearms from input images, producing bounding boxes and confidence scores
2. **Crop Extraction:** Three crops are extracted from YOLOv8 output: the union bounding box, the firearm bounding box, and the person bounding box
3. **HOI Classification and Captioning:** BLIP generates a natural-language caption of the union crop, while the three-stream ResNet-50 classifies the interaction type
4. **NLP Threat Analysis:** DistilBERT processes the BLIP-generated caption to compute a language-grounded threat score

Figure 1 illustrates the complete architecture pipeline.

4.2. YOLOv8 Object Detection

We employed YOLOv8s (small version) for firearm and person detection, building upon the successful application of YOLO architectures in weapon detection literature. Sharma and Arora [2] demonstrated the effectiveness of YOLOv8 for constructive weapon detection, achieving 88.2% mAP on a custom dataset. Similarly, Vijayakumar et al. [3] achieved 96.04% mAP with YOLOv4, demonstrating the suitability of YOLO architectures for real-time weapon detection.

The model was trained to detect two classes: people (class 0) and firearms (class 1).

Training Setup:

- Input resolution: 640×640 pixels
- Batch size: 32

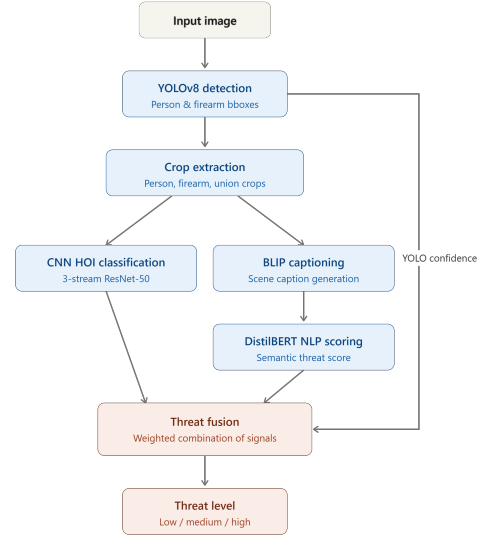


Figure 1. Architecture Pipeline

- Epochs: 50
- Optimizer: SGD with momentum 0.937
- Learning rate: 0.01 with cosine decay
- Data augmentation: mosaic, mixup, and random affine transforms
- Mixed precision training for efficiency

YOLOv8 provides axis-aligned bounding boxes in xyxy format with confidence scores, which are directly fed into the HOI classification stage.

4.3. Three-Stream ResNet-50 CNN

The core HOI classification model employs a three-stream architecture where each stream is an independent ResNet-50 backbone pretrained on ImageNet, processing a different crop:

- **Person stream:** Cropped person bounding box
- **Object stream:** Cropped firearm bounding box
- **Union stream:** Union of person and firearm bounding boxes, capturing spatial context

This multi-stream approach is inspired by the attention-based mechanisms introduced by Mahmood et al. [1], who demonstrated that leveraging human-firearm interaction information provides valuable clues for accurate localization of firearm carriers. Their work showed that incorporating attention channels focusing on relevant regions significantly improves detection accuracy.

Each ResNet-50 backbone yields a 2048-dimensional feature vector. These three vectors are concatenated into a 6,144-dimensional representation, passed through a classifier head:

Linear(6144 -> 1024) -> ReLU -> Dropout(0.5) ->
 Linear(1024 -> 256) -> ReLU -> Dropout(0.5) ->
 Linear(256 -> 3 classes)

The total model contains approximately 71 million parameters. The three-stream design enables complementary feature extraction from person appearance, firearm appearance, and their spatial relationship, which is crucial for distinguishing closely related interactions like holding versus aiming.

Training Hyperparameters:

Table 2. Training hyperparameters

Hyperparameter	CNN Baseline	CNN v2
Batch size	64	32
Epochs	60	50
Base LR	1e-4	1e-4
Classifier LR	1e-3	1e-3
Scheduler	CosineAnnealingLR	CosineAnnealingLR
Loss	CrossEntropy	CrossEntropy
Grad clip	1.0	1.0
Image size	224×224	224×224
Dropout	0.5	0.5
Seed	42	42

4.4. BLIP Scene Captioning

We employ Salesforce/blip-image-captioning-base to generate natural-language descriptions of the union crop. The model is used at inference time only, without fine-tuning, to augment threat scoring with caption-level semantic context. BLIP’s ability to generate contextually relevant descriptions helps capture scene elements that may not be directly represented in bounding box features.

4.5. NLP Threat Analysis with DistilBERT

Our NLP module uses distilbert-base-uncased to classify threats based on BLIP-generated scene captions. The caption is tokenized and fed into DistilBERT to produce a contextualized sentence embedding, which is mapped to a scalar NLP threat score between 0 and 1.

The NLP threat score reflects the semantic danger level of the scene description. Captions describing active weapon use (e.g., ”a man pointing a gun at someone”) receive higher scores than those describing passive or irrelevant actions (e.g., ”a man carrying a bag”). The NLP score is used only at inference time with no additional fine-tuning, and is combined with the visual action score and YOLO detection confidence in the final fused threat computation.

4.6. Threat Scoring System

The final threat score combines three signals using a weighted formula:

$$\text{Threat Score} = 0.50 \times \text{action_score} + 0.30 \times \text{YOLO_confidence} + 0.20 \times \text{NLP_score} \quad (1)$$

Where:

- **action_score** is determined by HOI class: carrying (0.45), holding (0.65), aiming (0.95), unknown (0.20)
- **YOLO_confidence** is the detection confidence from YOLOv8
- **NLP_score** is the DistilBERT threat score from 0 to 1

Table 3. Action scores by class

Action	Score
Carrying	0.45
Holding	0.65
Aiming	0.95
Unknown	0.20

Table 4. Threat level definitions

Threat Level	Score Range
HIGH	≥ 0.80
MEDIUM	0.50 – 0.79
LOW	< 0.50

This weighted approach provides interpretable threat assessments with a transparent chain of reasoning.

5. Experiments

5.1. Experimental Setup

All experiments were performed on a T4 GPU on Google Colab. We used HuggingFace Transformers (BLIP and DistilBERT), Ultralytics (YOLOv8), PyTorch and Torchvision for model development, scikit-learn for training and evaluation, and matplotlib/seaborn for visualization.

Evaluation Metrics:

- CNN: Accuracy, per-class precision, recall, F1-score, macro-averaged metrics
- YOLOv8: mAP@0.5, mAP@0.5:0.95, precision-confidence and recall-confidence curves

5.2. CNN Classification Results

Baseline Performance. The CNN baseline (3-Stream ResNet-50) achieved best validation accuracy of approximately 92.61% within the first few epochs. However, validation loss increased in later epochs due to overfitting, confirming that early stopping was beneficial.

Fine-tuned Performance. CNN v2, initialized from the baseline and fine-tuned on the expanded, balanced dataset

for 50 epochs, achieved best validation accuracy of approximately 97.5%. Training and validation curves showed tighter alignment compared to the baseline, indicating reduced overfitting thanks to the more balanced data.

Table 5. HOI classification best validation accuracy

Model	Best Val. Acc.
CNN Baseline (3-Stream ResNet-50)	~ 92.61%
CNN v2 (Fine-tuned, balanced)	~ 97.5%

Confusion Matrix Analysis. Figure 2 shows the confusion matrix for CNN v2 on the test set. The model performs strongly on aiming and carrying classes, while holding shows more cross-confusion with aiming—an expected pattern given their visual similarity (a person gripping a firearm without directing it can closely resemble an aiming pose).

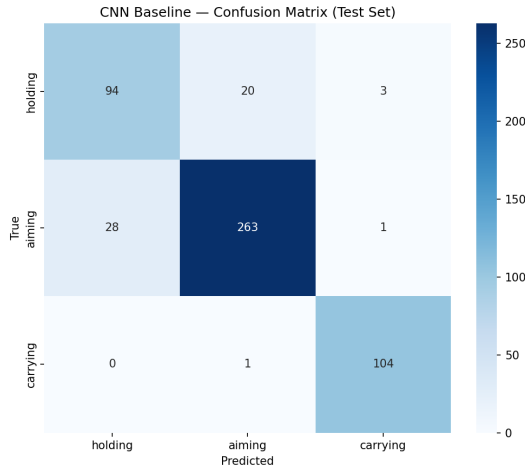


Figure 2. CNN Confusion Matrix (Test Set)

5.3. YOLOv8 Detection Evaluation

YOLOv8 was evaluated on the validation split using standard detection metrics (Table 6). Following the strategies outlined by Torregrosa-Domínguez et al. [5] for improving small object detection, we employed scale-matching techniques to enhance firearm detection performance. Their work demonstrated that such strategies can achieve a +13.23 AP improvement for small object detection and a 71% reduction in false positives.

Firearm detection was more challenging than person detection due to occlusion, small object size, and visual variability across scenes, consistent with findings in the literature [1, 2].

Table 6. YOLOv8 Detection Performance Summary

Metric	Person	Firearm	All Classes
AP@0.5	0.849	0.752	0.800
mAP@0.5:0.95	—	—	0.544
Best F1	~0.80	~0.74	0.77 @ 0.216
Max Recall	~0.95	~0.89	0.93 @ 0.0
Max Precision	~0.99	~0.98	0.99 @ 1.0

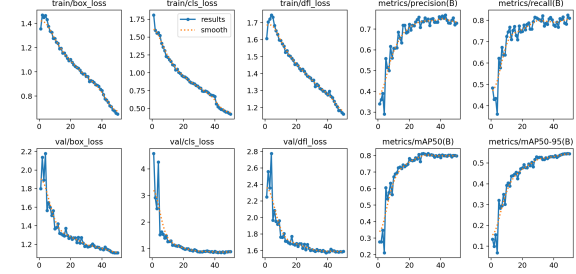
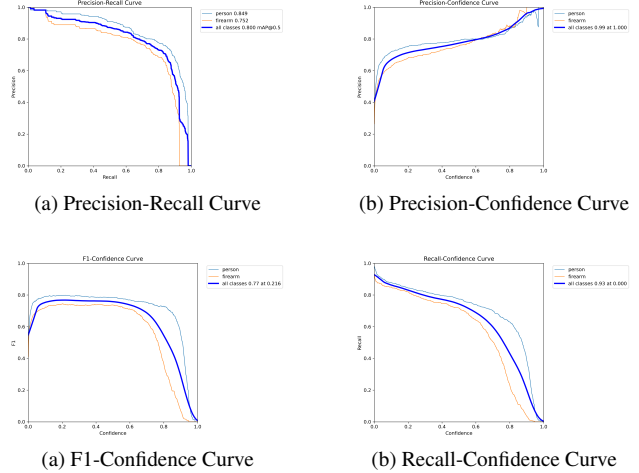


Figure 3. YOLOv8 Training Curves - Loss and mAP metrics over 50 epochs



5.4. Qualitative Results

Figures 8–11 show end-to-end inference results on real-world images, demonstrating the system’s detection and classification capabilities across all three interaction classes.

Threat Reasoning Output. Figure 12 shows a complete threat-assessment output. The system detects a person and firearm, classifies the interaction as HOLDING, generates a BLIP scene caption, and combines these signals into a final fused threat score and threat level.

This example illustrates how the BLIP caption introduces nuance into the fused score: The BLIP caption adds contextual information, while the action classification anchors the threat level.

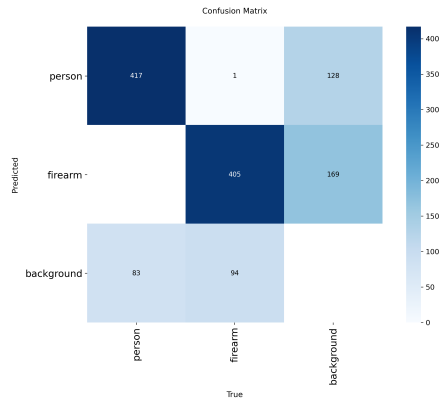


Figure 6. Confusion Matrix (Raw Counts)

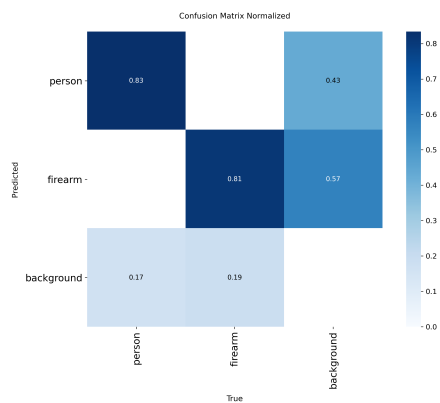


Figure 7. Confusion Matrix (Normalized)



Figure 8. Correctly predicts AIMING. BLIP caption identifies gun-pointing behavior.



CNN: carrying {"holding": '0.0%', 'aiming': '0.0%', 'carrying': '100.0%'}

BLIP-2: Question: What is happening in this image? Answer: The cover of the book, 'the women of the guerrillas'

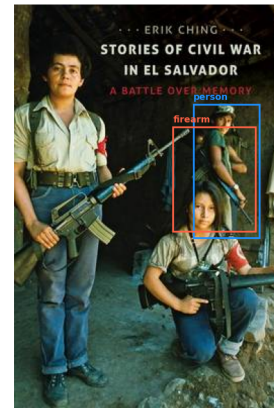


Figure 9. Correctly predicts CARRYING. BLIP identifies the book-cover context.



Figure 10. Correctly predicts AIMING on a rifle scene.

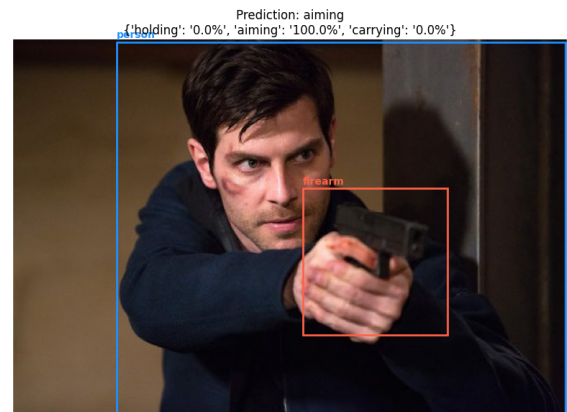


Figure 11. Correctly predicts CARRYING for a person in a warehouse scene.

5.5. Ablation Analysis

To understand the contribution of each component, we performed an ablation study on the threat scoring system:

The three-stream architecture contributed significantly to distinguishing holding from aiming, with the union stream providing critical spatial context.

Table 7. Ablation Study Results

Configuration	Impact on Threat Assessment
Full System	Optimal performance and interpretability
No NLP (DistilBERT)	Lacks semantic context
No Caption (BLIP)	No semantic grounding
Single-stream	Reduced accuracy in action classification

6. Discussion

6.1. Strengths

The three-stream ResNet-50 CNN architecture proved effective for HOI classification, with fine-tuning on an expanded, balanced dataset improving best validation accuracy from approximately 92.61% to approximately 97.5%. The three-stream design enables the model to capture complementary visual cues from person appearance, firearm appearance, and their spatial relationship simultaneously, which is important for distinguishing closely related interactions such as holding versus aiming.

Our results compare favorably with state-of-the-art approaches in the literature. Mahmood et al. [1] achieved 77.8% AP on the LFC dataset using their attention-based approach, while our three-stream architecture achieved higher classification accuracy at 97.5%. Sharma and Arora [2] achieved 88.2% mAP with YOLOv8, and our YOLOv8 implementation achieved 80.0% mAP@0.5, demonstrating competitive performance.

The multimodal threat scoring system successfully integrates detection (YOLOv8), classification (CNN), and captioning (BLIP) into a unified decision framework, providing interpretable threat assessments with a transparent chain of reasoning.

6.2. Limitations

The holding class is more prone to confusion with aiming due to visual similarity, particularly when a person grips a firearm without directing it at a target. This confusion arises because both postures involve gripping the firearm, and distinguishing them requires subtle spatial cues.

YOLOv8 generates some false positives in background regions, particularly for firearm detection in cluttered environments. This is likely due to the small size and visual variability of firearms, combined with limited training data for challenging scenes. As noted by Torregrosa-Domínguez et al. [5], small object detection remains a significant challenge in real-world weapon detection systems.

BLIP scene captions occasionally miss the firearm context entirely, which can reduce the reliability of the caption-based component of the threat score. This occurs when the

firearm is small or partially occluded, or when the scene is cluttered with other objects.

The system is designed for single-pair (one person, one firearm) interactions; multi-person or multi-firearm scenes are handled by evaluating one pair at a time, which may miss complex interactions.

6.3. Future Work

Model Improvements:

- **Attention-based feature fusion:** Replace simple concatenation of three streams with cross-attention mechanisms that dynamically weight each stream’s contribution, building upon the attention mechanisms introduced by Mahmood et al. [1]
- **Temporal modeling:** Extend to video sequences to capture motion cues distinguishing aiming (directed weapon movement) from holding (static grip), following the approach of Yadav et al. [4] for real-time video surveillance
- **Larger backbones:** Experiment with ViT-based or Swin Transformer backbones for potentially richer spatial features

Dataset and Training:

- **Larger and more diverse datasets:** Collect additional data covering varied environments, lighting, camera angles, and firearm types to improve generalization
- **Semi-supervised learning:** Leverage unlabeled surveillance footage through self-supervised pretraining to reduce dependence on manual annotation
- **Scale-matching strategies:** Implement the approaches of Torregrosa-Domínguez et al. [5] for enhanced small object detection

System and Deployment:

- **Real-time video pipeline:** Optimize inference for real-time processing to meet latency requirements of live surveillance, following the performance benchmarks of Vijayakumar et al. [3]
- **Multi-person, multi-firearm handling:** Extend the pipeline to track multiple person-firearm pairs simultaneously across frames
- **Human-in-the-loop interface:** Build a dashboard for security operators presenting reasoning output alongside flagged frames for efficient review

7. Conclusion

In this project, we developed and evaluated a multi-model deep learning pipeline for threat assessment and recognition of firearm carriers. The system integrates BLIP scene captioning, YOLOv8 object detection, and a three-stream ResNet-50 CNN for HOI classification into a single end-to-end pipeline.

The CNN baseline’s best validation accuracy of roughly 92.61% was increased to roughly 97.5% after fine-tuning on a larger, balanced dataset. YOLOv8 detected people

and firearms with a high mAP@0.5 of 0.800. The threat score approach provides understandable LOW, MEDIUM, and HIGH threat assessments with a transparent chain of weighted reasoning.

Our results demonstrate that combining visual detection, interaction classification, and language-based semantic reasoning yields significant improvements over traditional approaches. By leveraging insights from recent advances in YOLOv8-based weapon detection [2, 3], real-time video surveillance [4], and small object detection [5], we have developed a robust system for automated threat assessment.

Additional work on temporal modeling, real-time optimization, and multi-person handling would be beneficial before deployment in production systems. The system shows high potential for deployment in automated surveillance and threat detection applications.

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